

Continuity: Scope Dossier

The research case for a strategy-continuity intelligence layer at quant trading firms, the frontier capabilities and objection kill-sheet, and the product this build ships. Three parts, self-contained: read Part One for the problem and the Jane Street case, Part Two for the extended capabilities and defenses, Part Three for what we are building this weekend.

Prepared 22 August 2026 · Supersedes the separate Strategy Continuity paper and Frontier Addendum by combining them · Citation codes resolve inside each part; the full 130+ source trail lives in the companion Research Dossier · Material marked **TEAM** is the team's concept statement; **BRIEF** is the track brief; neither is evidence.

The Problem and the Case

1 · The concept, as stated by the team

TEAM CONCEPT STATEMENT **TEAM** (FROM TEAM CHAT, 22 AUG 2026, LIGHTLY FORMATTED)

"An intelligence layer on employee arrival across trading desks and floors, to ensure strategies unique to traders deployed at firms stay within firms upon employee departure."

"Which will in turn be used to found an institution within these firms to continue to foster growth for graduates and new employees to be in sync with the firm's current, recent and former strategies too."

"The build is the foundation, and then we do a vertical plan as part of our pitch, backed by the research."

Restated as a product thesis: quant firms pay enormous sums to acquire talent and the strategies that talent creates, but the firm's ownership of those strategies is mostly legal fiction backed by non-competes, not operational reality backed by systems. The knowledge that makes a strategy work (why parameters are set where they are, which experiments failed, which data quirks the code silently routes around) lives in the researcher's head and nowhere else. When the person leaves, the firm keeps the code and loses the strategy. We propose a research-decision capture layer that runs from an employee's first day, making the strategy's reasoning a firm-owned asset continuously rather than in a panicked exit interview, and, long term, an internal institution that uses the same captured corpus to onboard graduates into the firm's current and former strategies.

2 · Problem statement

Quantitative trading firms face a structural asymmetry: their most valuable asset is tacit knowledge, and every mechanism they use to protect it operates **after** the knowledge has already concentrated in one or two heads. The industry's actual toolkit is restrictive covenants (non-competes now reaching two years even for analysts [T4]), garden leave paid at five figures a month [T5], and litigation [T1, T6]. None of these transfer knowledge; they only delay its exit. Meanwhile churn at the pod-shop end of the industry runs at 15-20% of portfolio managers per year, with three-year retention near 55% [T8, T9], so the average strategy will outlive its author's tenure at the firm.

The problem has a mirror image that makes it worse: because strategy reasoning is undocumented, firms also cannot see changes happening inside their own models. The SEC's 2025 order against Two Sigma records an employee making unauthorized changes to more than 14 live models over roughly two years without detection, vulnerabilities flagged internally in 2019 and unremediated until 2023, at a cost of \$90M in penalties plus \$165M voluntarily repaid [O2, T24]. Capture and change-visibility are the same missing system viewed from two sides: the firm neither knows what its people know, nor notices when what they built quietly changes.

The track brief identifies this direction ("when a quant leaves, their undocumented knowledge often leaves with them") **BRIEF**; this paper's contribution is showing the problem is documented in primary sources strong enough to build on without any interviews: a federal

court filing, two SEC orders, industry turnover data, and two working precedents proving structured capture is possible and profitable.

3 · The evidence that this is real

THE NUMBERS THAT CARRY THE PITCH

- **>50% profit drop in one month after two departures**, on a strategy that made about \$1B in 2023 and was built by roughly 25 people over years: Jane Street's own amended complaint against Millennium (S.D.N.Y., 2024) [T1, T2].
- **15-20% annual PM turnover; ~55% three-year retention** at the large multi-manager funds; Millennium described as cutting PMs at 7.5% drawdowns [T8, T9].
- **\$90M penalty + \$165M repaid**: Two Sigma, for model changes nobody could see (SEC, 2025) [O2].
- **\$20M**: what Millennium paid Renaissance in 2007 over two hired quants, the same defendant seventeen years before the Jane Street case [T6].
- **Up to 2-year non-competes even for analysts** (Citadel, reported Aug 2026), and firms paying \$15-18K per month to enforce garden leave [T4, T5].
- **The industry admits it on the record**: the hedge fund lobby (MFA) formally argued to the FTC that non-competes are essential because trading IP leaves with people [T12].
- **Structured capture works and prints money**: Marshall Wace's TOPS turned structured idea submission into roughly \$30B of AUM [T14]; WorldQuant's BRAIN platform makes alpha research machine-legible by construction, with 650+ external consultants submitting through it [T13].

Three further facts shape the design. First, the knowledge really is tacit: the leading academic study of the industry, built on 300+ insider interviews, documents that trading knowledge is socially embedded and passed person to person, not through documents [T16]; Jane Street's own podcast concedes internal documentation never keeps up [F36]. Second, generic tools cannot reach it: enterprise knowledge search (Glean, \$7.2B valuation) indexes documents that exist, while quant knowledge lives in notebooks, parameter choices and failed-experiment memory that never becomes a document [T22, T23]. Third, firms already invest heavily in the adjacent, easier half: Man Group built and open-sourced ArcticDB so hundreds of quants share versioned research *data* [F20]; no equivalent exists for research *reasoning*. That asymmetry (data platforms everywhere, reasoning platforms nowhere) is the gap this concept fills.

4 · Application scenario: Jane Street, February 2024

This section grounds the product in the best-documented instance of the problem on the public record, so the pitch can reference a named firm with a court-filed account. Every factual claim below is from the amended complaint and settlement coverage [T1, T2, T3]; the counterfactual is ours.

What happened, per the court record

Jane Street built an India options strategy over years, involving roughly 25 people, generating about \$1B in profits in 2023 and described internally as "incredibly valuable" and IP-sensitive. Two traders, Douglas Schadewald and Daniel Spottiswood, resigned in February 2024 and joined Millennium. Per the complaint, the strategy's profits fell more than 50% in March 2024, the month after they left, and a new market participant appeared using the same broker with identical execution constraints. Jane Street sued, alleging trade-secret misappropriation; the defense argued the traders took only their "experience and expertise." The case settled in

December 2024 on undisclosed terms [T2]. A regulator later put the franchise's scale on the record: SEBI's 2025 order describes about \$4.3B of India profits from January 2023 to March 2025 [T3].

Why this is the perfect reference case

The lawsuit's central ambiguity is precisely the product's reason to exist. Jane Street had the code, the infrastructure, and 23 of the 25 people, yet lost half the strategy's output in a month, and then could not cleanly demonstrate in court where "firm IP" ended and the traders' personal "experience and expertise" began. A firm running a continuity layer would be in a different position on both counts: operationally (the reasoning, market intuitions, and execution constraints around the strategy would exist as a structured, queryable corpus that the remaining 23 people could work from) and legally (a timestamped record of firm-hosted contributions is close to the strongest trade-secret documentation a plaintiff can have; courts weigh whether the firm took reasonable measures to protect the secret, and a capture system is such a measure by construction).

The counterfactual, stated carefully for the pitch

We do not claim the tool would have prevented the departures, and we do not claim the profits would have been untouched; skill is real and some knowledge cannot be serialized. The claim is narrower and defensible: the firm's recovery time and its legal position both improve with a decision-genealogy corpus, and the gap between "we kept the code" and "we kept the strategy" narrows. The complaint itself documents which knowledge mattered by fighting over it: models, signals, execution constraints, broker arrangements. Those are exactly the artifacts the layer captures.

The buyer-side scenario: Millennium as the high-churn profile

The same case shows the other customer. Millennium runs 300+ autonomous pods with 15-20% PM turnover and has now twice been the defendant in landmark quant knowledge disputes (2007 Renaissance, \$20M; 2024 Jane Street) [T6, T1, T8]. A firm whose operating model is deliberately high-churn is structurally dependent on knowledge surviving personnel change, and currently has no system that makes it do so. Two Sigma is the third scenario: its \$255M outcome was not about departure at all but about the absence of a record of who changed which model, when, and why [O2, T24]; the same capture layer is that record.

5 · What we build this weekend

Working name: **Continuity** (rename freely). The hackathon prototype is the intelligence layer's core loop, demonstrated on a realistic synthetic trading-research repo (strategy code, notebooks, parameter files, a fake chat log) that we author on Friday so the demo is credible without any proprietary data.

Prototype scope (the foundation)

- **Ambient capture.** Git hooks and a Jupyter extension that intercept every merged change to strategy code or parameters and require a one-line structured rationale (what changed, why, what was tried and rejected). Each entry auto-links the diff, author, timestamp, and affected strategy. This is the decision genealogy: the record whose absence cost Two Sigma \$255M [O2] and whose presence would have been Jane Street's evidence [T1].
- **The debrief agent.** An LLM interviewer that runs short scheduled sessions with each researcher ("you changed the vol filter threshold twice this month; walk me through why"), generating targeted questions from the capture stream rather than generic prompts, and

filing transcripts against the right strategy. Modeled on the proven structured-capture precedents [T13, T14] but aimed at reasoning, not ideas.

- **Knowledge concentration metrics.** A bus-factor dashboard per strategy: who is the sole author or toucher of which components, weighted by P&L attribution; a departure simulation showing what fraction of a strategy's attributed code and decisions would lose their last knowledgeable owner.
- **The handover pack.** One command generates, for any (simulated) departing employee, a structured succession document: their strategies, their decision history, the open questions only they can answer ranked by undocumentedness, and a ready-made interview script for their final weeks.

Demo script (three minutes)

Open on the Jane Street numbers: a \$1B strategy, two resignations, profits down more than 50% in a month, per the firm's own court filing [T1, T2]. Show the synthetic firm's dashboard: one strategy glowing red (bus factor 1). Simulate that person resigning. Generate the handover pack live; show the debrief agent conducting the exit session and the knowledge map turning from red toward green as answers file in. Close on the two-sided value: continuity for the desk, and a timestamped IP record the general counsel would have paid for in April 2024.

Why the research says this is the right build

SOURCE FINDING	DESIGN DECISION IT FORCED
Knowledge is tacit and never becomes documents [T16, F36]; enterprise search only indexes what exists [T22]	Capture at the moment of action (commits, parameter changes), not via wikis or after-the-fact documentation drives
Exit interviews and wikis demonstrably fail for tacit knowledge [T23]	Continuous ambient capture from day one TEAM , with the exit pack as output, not input
Structured capture succeeded when it was lightweight and incentive-aligned (TOPS, BRAIN) [T14, T13]	One-line rationales and short agent sessions, never long forms; visible credit for captured contributions
The Two Sigma failure was invisibility of change, not absence of talent [O2]	The same stream doubles as a model change-control audit trail, which also answers "why would a firm buy this"
Firms already accept research-infrastructure mandates (ArcticDB for data) [F20]	Position as research infrastructure, not surveillance; it sits beside the data platform as the reasoning platform

6 · Long-term vision and vertical plan

The long-term thesis, per the team concept **TEAM**: the capture layer's corpus becomes the foundation of an institution inside the firm, so that continuity stops being an emergency process and becomes how the firm teaches itself.

Phase 1 (hackathon to seed): the continuity layer.

The prototype above, hardened: on-prem or VPC-only deployment (no quant firm ships this data to a SaaS cloud), integration with git, Jupyter, and chat, and the bus-factor and change-control dashboards. Wedge buyer: the COO and general counsel, because the same artifact serves succession risk and trade-secret documentation. The reference sale writes itself from the litigation record cited above.

Phase 2 (the institution): onboarding built on the corpus.

Once decisions and debriefs accumulate, the corpus becomes courseware. New graduates onboard against the firm's actual current, recent, and former strategies **TEAM**: guided walkthroughs of a live strategy's decision history, "why is this parameter 0.7" answered by the recorded rationale rather than by interrupting the one person who knows. This attacks the documented onboarding gap (sink-or-swim is the industry's stated norm; documentation never keeps up [F36]) and shortens the time-to-productive that firms currently buy with two-year non-competes on the other end.

Phase 3 (the flywheel): garden leave and alumni knowledge.

Firms pay departing quants five figures monthly to sit idle for up to two years [T4, T5]. A compliance-safe, firm-side structured debrief program during that paid window converts dead spend into curriculum, and the corpus becomes the firm's institutional memory of strategies past: searchable, teachable, and legally owned. At this point the "institution within the firm" exists in fact: capture feeds curriculum, curriculum feeds new researchers, whose work is captured in turn.

Why now. Three trends converge: non-compete law is under active attack (the FTC's attempted ban and state-level movement [T12]), which weakens the legal substitute for actual knowledge retention; LLMs make the debrief agent and corpus querying feasible for the first time; and regulators are converging on the same artifact from the compliance side (model change-control expectations post-Two Sigma [O2], algo inventories and documented change management under MiFID II RTS 6 [O10, O11]). The product's record is simultaneously a retention asset, a legal asset, and a compliance asset.

7 · Objections we will face, answered honestly

- **"Traders will refuse to feed it."** True if it reads as surveillance or as making them replaceable. Mitigations built into the design: capture is about strategies, not people-monitoring; contribution is visible and creditable (the TOPS lesson: people feed systems that pay them credit [T14]); friction is kept to one line. We say this openly in the pitch; pretending incentives away would be the weak move.
- **"Tacit knowledge cannot be fully captured."** Correct, and the pitch never claims total capture. The claim is marginal: the difference between profits falling 50% and falling meaningfully less, and between an unprovable trade secret and a documented one. The court record shows the marginal value is measured in hundreds of millions [T1, T2].
- **"Firms would build this in-house."** The strongest objection, since firms build everything [F18-F22]. Two answers: they have not (the confessed tooling for data exists, for reasoning does not), and the compliance-artifact framing gives an external product a neutral-record advantage an internal tool lacks in litigation.
- **"Is this just a wiki with hooks?"** No: wikis wait for voluntary prose after the fact and demonstrably fail [T23]. This captures at the point of action, interrogates actively, and measures concentration. The deliverable is a genealogy and a risk metric, not pages.

8 · Pitch reference card and citations

Every claim in the pitch or video should be spoken with its source. The quotable core, then the cited subset in full (codes match the main Research Dossier).

SAY THIS	BACKED BY
"A \$1B-a-year strategy lost over half its profits in the month after two people resigned. That is Jane Street's own number, in its own federal court filing."	Amended complaint, S.D.N.Y. 2024 [T1]; settlement coverage [T2]

"The multi-manager funds turn over 15 to 20 percent of their portfolio managers every year. The average strategy will outlive its author's seat."	Net Interest, "Peak Pod" [T8]; Hedgeweek/WSJ [T9]
"Two Sigma paid \$90M and repaid \$165M because nobody could see who changed 14 live models over two years. The missing system is the same one."	SEC press release 2025-15 [O2]; MoFo alert [T24]
"The industry's own lobby told the FTC that non-competes are essential because strategies leave with people. That is the industry admitting it has no operational answer, only a legal one."	FTC record and MFA opposition [T12]
"Structured capture is proven: Marshall Wace turned it into roughly \$30B of AUM, and WorldQuant industrialized it into a platform."	Net Interest, "Alpha Capture" [T14]; WorldQuant BRAIN [T13]
"Firms built world-class platforms for research data. Nobody has built one for research reasoning."	ArcticDB and peers [F20, F18, F19]; gap analysis, Dossier 2.7

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- T2** · Settlement coverage (Bloomberg-sourced), Dec 2024; ~\$1B 2023 profits on the disputed strategy. <https://www.claimsjournal.com/news/national/2024/12/09/327780.htm>
- T3** · SEBI interim order coverage, July 2025: ~\$4.3B India profits Jan 2023-Mar 2025. Primary PDF retrievable from [sebi.gov.in](https://blogs.law.ox.ac.uk/oblb/blog-post/2025/07/jane-street-and-expiry-day-trap-unpacking-sebis-crackdown-algorithmic). <https://blogs.law.ox.ac.uk/oblb/blog-post/2025/07/jane-street-and-expiry-day-trap-unpacking-sebis-crackdown-algorithmic>
- T4** · Bloomberg, Aug 2026: Citadel's 1-2 year non-competes down to analyst level (paywalled; corroboration linked). <https://www.briefs.co/news/citadel-s-two-year-ban-on-job-moves-stalls-analyst-careers/>
- T5** · Institutional Investor, 2022: \$15-18K/month garden-leave compliance payments, Citadel v. Prieur. <https://www.institutionalinvestor.com/article/b1zyvmvxf2612/An-Employee-Left-Citadel-Securities-for-a-Startup-Now-He-s-Facing-a-Lawsuit>
- T6** · Renaissance v. Millennium: \$20M settlement over two hired quants (2007), confirmed by counsel. <https://www.cravath.com/renaissance-technologies-corp-settles-trade-secrets-dispute/>
- T8** · Rubinstein, "Peak Pod," Net Interest: 15-20% annual PM turnover; ~55% three-year retention; Millennium ~300 teams. <https://www.netinterest.co/p/peak-pod>
- T9** · Hedgeweek (WSJ-derived), 2025: nine-figure packages; PMs cut at 7.5% drawdowns. <https://www.hedgeweek.com/multi-manager-hedge-funds-escalate-talent-war-arms-race-with-nine-figure-pay-packages/>
- T12** · FTC non-compete rule (2024) and MFA opposition: the industry's on-record argument that trading IP leaves with people. <https://www.ftc.gov/news-events/news/press-releases/2024/04/ftc-announces-rule-banning-noncompetes>
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- T14** · Rubinstein, "Alpha Capture," Net Interest: Marshall Wace TOPS, ~500,000 structured ideas/year by 2005, ~\$30B AUM. <https://www.netinterest.co/p/alpha-capture>
- T16** · MacKenzie, "Trading at the Speed of Light" (Princeton, 2021): 300+ insider interviews; trading knowledge is tacit and socially embedded. <https://press.princeton.edu/books/hardcover/9780691211381/trading-at-the-speed-of-light>
- T22** · Glean (enterprise search, \$7.2B valuation) indexes existing documents; quant reasoning never becomes one. https://en.wikipedia.org/wiki/Glean_Technologies
- T23** · HR literature on wikis and exit interviews failing to capture tacit knowledge. <https://hrexecutive.com/unshared-knowledge-destroys-a-companys-bottom-line/>
- T24** · Morrison Foerster on the Two Sigma order: no systematic record of who changed which model parameter, when, and why. <https://www.mofo.com/resources/insights/250226-top-5-sec-enforcement-developments>

- O2** · SEC press release 2025-15, Two Sigma: unauthorized changes to 14+ live models; \$90M penalty, \$165M repaid; flagged 2019, fixed 2023. <https://www.sec.gov/newsroom/press-releases/2025-15>
- O10** · FCA multi-firm review, 2018: firms could not inventory their own algorithms; limited dev/test/deploy documentation. <https://www.fca.org.uk/publication/multi-firm-reviews/algorithmic-trading-compliance-wholesale-markets.pdf>
- O11** · Central Bank of Ireland thematic review, 2021: RTS 6 self-assessments deficient; documentation of controls lacking. <https://www.centralbank.ie/docs/default-source/regulation/industry-market-sectors/investment-firms/mifid-firms/regulatory-requirements-and-guidance/thematic-assessment-of-algorithmic-trading-firms-compliance.pdf>
- F18** · HRT, distributed filesystem for research at hundreds of petabytes. <https://www.hudsonrivertrading.com/hrtbeat/distributed-filesystem-for-scalable-research/>
- F19** · Two Sigma, "Treating Data as Code." <https://www.twosigma.com/articles/treating-data-as-code-at-two-sigma/>
- F20** · Man Group, "Our Journey Creating ArcticDB": the research data platform firms do build; no reasoning equivalent exists. <https://arcticdb.io/blog/our-journey-creating-arcticdb-solving-the-challenge-of-dataframes-at-scale/>
- F22** · G-Research, Armada: research compute platform, open-sourced. <https://armadaproject.io/>
- F36** · Jane Street, Signals and Threads, "Writing, Technically": internal documentation never keeps up. <https://signalsandthreads.com/writing-technically/>

Integrity notes for the video: the >50% figure is Jane Street's allegation in a settled case; say "per Jane Street's court filing," not "proven in court." Millennium's 7.5% drawdown rule and turnover figures are industry reporting, not filings. The Two Sigma numbers are from the SEC's own press release and are safe to state plainly.

The Frontier Build and the Objection Kill-Sheet

1 · What this addendum found, in one paragraph

Three discoveries reshape the pitch. First, the friction objection is solvable with published patterns: provenance tools that capture with literally zero workflow change exist as open source [C1-C6], and the research shows the real bottleneck was never writing effort but deciding what to record [C19], which is exactly what an LLM agent automates. Second, regulators have already written our product spec: the FCA criticized firms for taking "key decisions... without sufficient explanation of the underlying rationale" [L7], the Fed defines model documentation as detailed enough that "parties unfamiliar with a model" could understand it [L9], and UK handover rules literally require material containing "judgement and opinion, not just facts and figures" [I17]. Third, the strongest new fact in the whole sweep is legal: US courts dismiss trade-secret cases, regardless of whether theft actually happened, when the firm cannot describe its secret with "reasonable particularity" [L2], and roughly 11% of cases die on the "reasonable measures" requirement alone [L1]. An undocumented strategy may be legally undefendable. That upgrades Continuity from a nice-to-have into the thing that determines whether the Jane Street lawsuit is even possible.

2 · The upgraded architecture

Version 2 of the product is five layers. The weekend build implements slices of layers 1, 3 and 5 (Section 6); the rest is the vision, now grounded item by item.

LAYER	WHAT IT DOES	GROUNDING
1 Capture	Zero-friction provenance: every run, diff and parameter change recorded as a side effect of working; LLM drafts the rationale, the human approves	noWorkflow, Verdant, papermill pattern [C1, C2, C5]; Copilot PR summaries [C8]
2 Firm brain	A knowledge graph over the firm's exhaust (code, chat, notebooks) answering global questions with citations	Microsoft GraphRAG [G1]; rationale-mining field [G3, G4]
3 Elicitation	The debrief agent: LLM interviewer that detects significant decisions, asks targeted questions at natural pauses, schedules follow-ups before memory decays	LLMREI [G11]; Duolingo half-life regression [G13]; Morgan Stanley Debrief [G14]
4 Continuity assets	Handover packs, onboarding courseware, warm-start priors for successors, and (the headline) the researcher twin	SYSC 25.9 handover spec [I17]; AutoElicit [G12]; Stanford generative agents [G6]
5 Proof and risk	Cryptographic timestamping and tamper-evident logs; knowledge-at-risk metrics (bus factor, vacation-readiness) and auto-drafted compliance artifacts	RFC 3161, Merkle logs [G17, G18]; truck-factor algorithm [G21]; RTS 6, SR 11-7 [L6, L9]

3 · The frontier capabilities, each with its honest limit

3.1 Zero-friction capture (kills the friction objection)

The academic record shows capture can be invisible. noWorkflow collects full provenance of Python research code requiring "neither changes to the script, nor any laborious configuration" [C1]; Verdant (a CMU best-paper) records every notebook cell and output edit passively in the

background [C2]; nbsafety tracks cell-level lineage from a drop-in kernel, and the cells it flagged were seven times more likely to be what users actually re-ran [C4]; Netflix made every notebook run self-documenting by construction with papermill's immutable output artifacts [C5]; and JupyterLab ships an official telemetry extension point, so none of this requires forking the tools quants already use [C6]. On the writing side, LLMs already beat state-of-the-art at generating the "what changed and why" record from a diff [C7], and GitHub ships PR summaries commercially, with a documented rule we adopt: the model drafts, the human approves, because hallucinated rationale is worse than none [C8]. The design also respects an empirical constraint: developers switch activities every 0.3 to 2.0 minutes [C15], so Continuity never interrupts mid-flow; it batches approvals at natural boundaries (merge, promotion to paper trading), which is where the ADR field-study says documentation actually succeeds [C18, C19]. *Limit: capture of artifacts is total; capture of reasoning still depends on the human approving or correcting drafts.*

3.2 The firm brain (kills the "just a wiki" objection at scale)

Microsoft's GraphRAG builds an entity knowledge graph from source documents and pre-summarizes communities of related entities, enabling global questions like "what do we collectively know about crowding in momentum?" answered with citations [G1]; it is open source. Repository-level code question-answering is now a benchmarked research task (576-question SWE-QA spanning "intention understanding, cross-file reasoning, and multi-hop dependency analysis") [G2], and mining decision rationale from commit histories is an established research field with current tooling (CoMRAT detects decision and rationale sentences in commit messages) [G3, G4]. The who-knows-what layer has twenty years of precedent: Expertise Browser computed expertise maps from change history at Bell Labs in 2002 [G24]. *Limit: graph indexing is LLM-expensive, quant jargon needs a custom extraction schema, and hallucinated graph edges must be provenance-checked against the underlying artifacts.*

3.3 Active elicitation, scheduled like spaced repetition

The 1980s named this exact problem: the "knowledge acquisition bottleneck" was the blocker that killed expert systems [G10]; the pitch line is that LLMs are the first credible attack on a forty-year-old bottleneck. Evidence they work as interviewers: an LLM requirements-elicitation interviewer made a similar number of errors to human interviewers and generated contextually appropriate follow-ups [G11]. Morgan Stanley already ships an AI meeting "Debrief" agent to its advisors, with adoption above 98% of advisor teams for its knowledge assistant [G14], proving the pattern lands in regulated finance. Our twist: schedule the questions with half-life regression (the Duolingo spaced-repetition model [G13]), so the agent asks about a decision just before organizational memory of it decays, and only when decision-detection heuristics fire (a threshold changed after a drawdown, a branch abandoned after N backtests). *Limit: elicitation quality in quant research specifically is unstudied, and interviewees can satisfice; mitigated by grounding every question in a concrete captured artifact.*

3.4 The researcher twin (the headline, stated honestly)

Stanford and DeepMind built generative agents of 1,052 real people from a two-hour structured interview each; the agents answered social-survey questions at about 85% of the accuracy of the person's own two-week test-retest consistency [G6]. Applied here: a departing researcher's exit debrief plus their captured decision corpus becomes a queryable agent; the successor asks "why did you cap position size in the expiry window?" and gets the answer the departed quant would have given, linked to the records that support it. Commercial precedent exists (Delphi's "digital minds") [G9], and captured beliefs can do more than answer questions:

AutoElicit shows elicited expert knowledge can be converted into Bayesian priors that warm-start the successor's models with fewer labels [G12]. State the limits on stage, because they are provable: imitation-learning theory shows clone error compounds off the expert's beaten path [G7], and Polanyi's tacit-knowledge problem ("we know more than we can tell") bounds what any corpus contains. The twin is a memory prosthesis for the desk, not a replacement trader; a finance-specific cloning attempt exists and regime-shift fragility is its known weakness [G8]. If the firm fine-tunes twin models, instructional fingerprinting lets it prove model ownership even after further fine-tuning [G20].

3.5 The cryptographic proof layer (turns the record into a legal weapon)

Every captured decision gets an RFC 3161 trusted timestamp (the IETF standard, with ANSI X9.95 as its financial-grade extension) [G17], chained into a Merkle-tree tamper-evident log of the kind that runs Certificate Transparency [G18]; OpenTimestamps provides free git-integrated anchoring implementable in an afternoon [G19]. Why this matters legally: contemporaneous records "kept in the course of a regularly conducted activity" are exactly what the business-records exception to hearsay admits into evidence [L20]; US broker-dealer rules already require records in tamper-evident form with a designated third party able to hand them to regulators [L19], which is direct precedent for "neutral third-party record" being a feature rather than a bug; and the DTSA case law (next section) makes a precisely-documented secret the difference between winning and being dismissed. *Limit: timestamps prove when and what, never who truly originated an idea; courts weigh them as strong supporting evidence, not a silver bullet.*

3.6 Knowledge risk, quantified like the firm quantifies everything else

The bus factor is a real algorithm, not a metaphor: published methods compute it from git history (46% of popular GitHub projects have truck factor 1) with an open-source tool to reuse [G21]; a practitioner survey found 75% rate it important, 63% hit it in the last year, only 19% ever saw it measured, and one respondent reported six person-months recovering from a single departure [I19]. Turnover-induced knowledge loss has been modeled as a rare-severe-event risk (the framing lets us pitch "knowledge VaR" to a risk-fluent audience) [G22], and Microsoft's ownership research shows concentration measurably predicts defects [G23]. The regulatory hook is delightful: finance already mandates a bus-factor fire drill. FINRA guidance and NY DFS letters require traders to take about two consecutive weeks of leave with access cut off, precisely because misconduct "requires the continued presence of the wrongdoer" [I18]. Continuity ships a **vacation-readiness score**: whether the desk passes its mandated two-week absence cleanly. *Limit: authorship metrics count who touched things, not who could rebuild them; the debrief corpus is what closes that gap.*

4 · The objection kill-sheet

Each objection from the Strategy Continuity paper, now answered with researched mechanisms rather than assertions. This table is built to survive judge Q&A.

OBJECTION	THE RESEARCHED ANSWER
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"Traders will refuse to feed it" (hoarding is rational)	Four stacked mechanisms. (1) Pay for it: Marshall Wace's TOPS paid contributors by measured idea performance (over \$250M/yr by 2004-05) and grew to ~\$30B of AUM; WorldQuant pays external consultants per alpha, top tiers earning \$8,000+ quarterly [16, 17, 18]. Continuity's decision records become P&L-attributed "idea tickets" wired to bonus pools. (2) Make the record the trader's asset: employers demonstrably trust third-party-verified contribution traces ("GitHub is your new resume") [113], so a firm-signed decision record is comp-negotiation ammunition; documentation becomes leverage, not its loss. (3) The research says fear of losing knowledge-power is empirically weaker than assumed, intrinsic motivators and visible credit dominate (badge introductions raised contribution by a median 88.6%, strongest among senior users) [11, 111, 112], and hoarding measurably backfires on the hoarder via distrust loops that reduce their own creativity [13]. (4) The dilemma has a known game-theoretic fix: restructure payoffs, show contributors their contribution mattered, make group identity salient [12]. Design rule from Bridgewater's cautionary tale: capture decisions, never rank-and-shame people; their recorded-everything culture worked as capture but the published league tables produced fear and churn [19, 110].
"It's surveillance"	Finance is the one industry fined for NOT recording: \$3B+ against 100+ firms since 2021 for failing to capture business communications [L12]; UK rules mandate recording even intent-stage conversations, retained five to seven years [L13]; trader voice transcription is installed infrastructure [C12]. Continuity structures what is already legally captured, and adopts the Microsoft Recall lesson for the boundary: opt-in, artifact-level (never screens), encrypted, transparent [C14]. In the EU the boundary is law, not taste: German works councils have co-determination over monitoring tech and a DPIA is required [L15, L14], so the product is architected as decision-record capture of work product, not behavioral analytics. The trader-side sweetener: WakaTime proves developers voluntarily run always-on telemetry when it gives them personal value back [C17]; every researcher gets their own searchable history first.
"Tacit knowledge can't be captured"	Conceded, and now bounded instead of hand-waved: imitation theory proves clones degrade off-distribution [G7], Polanyi's limit is real. But the addressable layer is large and the marginal value is enormous: the Stanford result shows a two-hour interview captures enough to hit ~85% of a person's own consistency on held-out questions [G6], elicited knowledge can warm-start successor models as priors [G12], and the legal system only requires the explicit layer anyway: trade-secret cases are won on documented particularity, not on tacit essence [L2].
"Firms would build it in-house"	For compliance-grade records, the world's most sophisticated banks overwhelmingly buy: Global Relay serves 22 of the top 25 banks, Smarsh 18 of the top 20 [L18]. The reason generalizes to us: US securities law itself requires a designated third party able to deliver records to regulators "even if the firm is unwilling or unable" [L19], and neutral, routinely-generated records carry evidentiary weight internal reconstructions lack [L20]. Continuity is the same buy logic one layer up: reasoning instead of messages.
"Just a wiki with hooks"	Wikis fail at the selection step, not the writing step [C19]; Continuity automates selection (decision detection), drafting (LLM from diffs [C7]), recall (graph queries with citations [G1]), scheduling (half-life regression [G13]), proof (timestamped tamper-evident log [G17, G18]), and measurement (truck factor, knowledge VaR [G21, G22]). No wiki does any of those six.
"Who is the buyer?"	Three, all above the trading floor. Legal: the general counsel, because ~11% of trade-secret cases die on reasonable measures and courts demand reasonable particularity the firm may not possess [L1, L2]; attorney commentary on <i>Jane Street v. Millennium</i> literally prescribes "document development and implementation meticulously" [L4]. Compliance: RTS 6 Article 5(7) requires a who/what/when/approved-by record of every material algo change, and SR 11-7 defines documentation as understandable by "parties unfamiliar with a model" [L6, L9]; the FCA has already cited firms for missing rationale [L7]. Capital: ILPA standards make a key person's departure an automatic suspension of the investment period, allocator DDQs demand documented succession plans, and half of hedge fund failures are operational, not investment, failures [L16, L14, L17]. The COO buys continuity, the GC buys evidence, IR buys DDQ answers; the trading floor just has to tolerate it, which is what Section 4 row one is for.

5 · The three "crazy but real" demo peaks

For the video and live pitch, in escalating order of audacity, each fully backed:

1. **The vacation-readiness score.** Show a desk failing its regulator-mandated two-week absence test before it happens [I18, G21]. One number, instantly legible to any finance judge, and no other team will have found the block-leave rules.
2. **The mock lawsuit.** Replay the Jane Street scenario against the synthetic firm: the departing trader's counsel claims "it was just his experience and expertise"; Continuity produces the timestamped, tamper-evident decision genealogy with reasonable particularity, and the slide cites the case law where its absence gets claims dismissed [L2, L1, G17]. This lands the triple-value framing in thirty seconds.
3. **Interviewing the departed quant.** The researcher twin answers the successor's question, citing its sources from the corpus, followed immediately by the honesty slide: what the twin can and cannot know, with the imitation-learning bound on screen [G6, G7]. Judges reward the audacity and remember the honesty.

6 · Weekend triage: what gets built vs pitched

CAPABILITY	WEEKEND VERDICT	NOTE
Capture hooks + LLM-drafted decision records (approve/edit flow)	Build	Core loop; all components proven [C1, C7, C8]
Bus-factor dashboard + departure simulation + vacation-readiness score	Build	Reuse the open-source truck-factor algorithm [G21]; the score is arithmetic on top
Debrief agent (session-end questions grounded in captured diffs)	Build	An LLM prompt over the capture stream; half-life scheduling can be stubbed [G11, G13]
OpenTimestamps anchoring of the decision log	Build (half a day)	Free, git-integrated; makes the mock-lawsuit demo real [G19]
Handover pack generator	Build	Template it on the SYSC 25.9 spec: "judgement and opinion, not just facts and figures" [I17]
Researcher twin	Demo-slice only	A RAG agent over the synthetic corpus in the departed persona is honest and impressive; do not claim fine-tuning [G6]
GraphRAG firm brain	Stretch goal	Open-source pipeline exists but indexing cost eats time; a small pre-built graph over the synthetic corpus is enough for one query on stage [G1]
Auto-drafted RTS 6 / SR 11-7 artifacts	One sample page	Generate a single filled self-assessment section from the capture stream; the citation does the rest [L6, L9]
Warm-start priors, model fingerprinting, comp integration	Pitch only	Vision slides with citations [G12, G20, I6]; claiming them as built would be dishonest and judges will probe

7 · Citations

Compact list; codes as used above. Quotes shown in Sections 1-6 are verbatim from fetched pages unless the entry notes otherwise. Items from paywalled or abstract-only sources are marked; verify before putting numbers on slides.

C · Capture and friction

- C1** · noWorkflow (VLDB 2017 + OSS): zero-config provenance for Python research code. <https://github.com/gems-uff/noworkflow>
- C2** · Verdant, CMU (VL/HCC 2018 best paper): passive background history of notebook experimentation. <https://marybethkery.com/Verdant/>
- C3** · ProvBook: per-cell execution provenance with diff UI. <https://ceur-ws.org/Vol-2180/paper-57.pdf>
- C4** · nbsafety (VLDB): drop-in kernel lineage; flagged cells 7x more likely to be re-executed. <https://arxiv.org/abs/2012.06981>
- C5** · Netflix papermill infrastructure: every notebook run emits an immutable, self-documenting output artifact. <https://netflixtechblog.com/scheduling-notebooks-at-netflix-348e6c14cfd6>
- C6** · JupyterLab telemetry (official): schema-based event hooks without forking. <https://github.com/jupyterlab/jupyterlab-telemetry>
- C7** · LLM commit-message generation study: superior to prior automated methods. <https://arxiv.org/abs/2404.14824>
- C8** · GitHub Copilot PR summaries + hallucination warning ("reviewing is a requirement"). <https://docs.github.com/en/copilot/responsible-use/pull-request-summaries>
- C12** · NICE Actimize trader voice transcription; Verint Capture: comms capture is installed infrastructure at trading firms. <https://www.niceactimize.com/financial-markets-compliance/trader-voice-transcription>
- C14** · Microsoft Recall backlash and redesign: off by default, authenticated, encrypted; the acceptability blueprint. <https://www.notebookcheck.net/Microsoft-makes-three-critical-changes-to-Windows-Recall-after-facing-backlash.845469.0.html>
- C15** · Meyer et al., IEEE TSE: developers stay 0.3-2.0 minutes per activity; interruption-based capture fails. <https://andre-meyer.ch/the-work-life-of-developers-activities-switches-and-perceived-productivity/>
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- C18** · Nygard, Architecture Decision Records (2011): the canonical lightweight rationale template. <https://www.cognitect.com/blog/2011/11/15/documenting-architecture-decisions>
- C19** · ADRs in Practice, ECSA 2024 action research: friction is deciding what to document, not writing it. https://rebekkaa.github.io/files/2024_ECSA.pdf
- C20** · LLM-generated ADRs (context-strategy evaluation): the debrief agent's academic precedent. <https://arxiv.org/html/2604.03826v1>

I · Incentives and adoption

- I1** · Kankanhalli, Tan, Wei (MIS Quarterly 2005): intrinsic motivators dominate; feared loss of knowledge-power empirically weak. <https://misq.umn.edu/contributing-knowledge-to-electronic-knowledge-repositories-an-empirical-investigation.html>
- I2** · Cabrera and Cabrera (Organization Studies 2002): sharing as a public-goods dilemma; fix payoffs, efficacy, identity. <https://journals.sagepub.com/doi/10.1177/0170840602235001>
- I3** · Cerne et al. (Academy of Management Journal 2014): knowledge hiding triggers distrust loops that reduce the hider's own creativity. <https://journals.aom.org/doi/10.5465/amj.2012.0122>
- I6** · Rubinstein, "Alpha Capture": TOPS paid by idea performance, >\$250M/yr by 2004-05, ~\$30B AUM today. <https://www.netinterest.co/p/alpha-capture>
- I7** · Alpha capture systems overview: ideas carry rationale, timeframe, conviction; performance is quantified per contributor. https://en.wikipedia.org/wiki/Alpha_capture_system
- I8** · WorldQuant BRAIN consultant program: paid per alpha quality; top tiers \$8,000+ quarterly. <https://worldquantbrain.com/consultant>
- I9** · HBS case 413-702, Bridgewater: total decision capture (videotaped meetings) at the world's largest hedge fund. <https://www.hbs.edu/faculty/Pages/item.aspx?num=44831>
- I10** · Bridgewater transparency backfire coverage: published rankings produced fear and churn; capture decisions, not people-rankings. <https://factually.co/fact-checks/business/bridgewater-radical-transparency-controversies-governance-61a95e>
- I11** · Bornfeld and Rafaeli (First Monday 2017): badge launches raised contribution, median +88.6%, strongest for committed users. <https://firstmonday.org/ojs/index.php/fm/article/download/7299/6301>

- I12** · Yanovsky et al. (JASIST 2021): gamified credit measurably steers effort; aim it at debrief quality, not volume. Abstract-level. <https://arxiv.org/pdf/2008.06125>
- I13** · Marlow and Dabbish (CSCW 2013): employers trust third-party-verified contribution traces; "GitHub is your new resume." <https://www.cs.cmu.edu/~xia/resources/Documents/Marlow-cscw13.pdf>
- I14** · AIMA Due Diligence Questionnaires: allocators' standard forms demand succession plans, deputies, turnover tables. <https://www.aima.org/sound-practices/due-diligence-questionnaires.html>
- I17** · FCA SYSC 25.9, handover procedures: material "should include judgement and opinion, not just facts and figures." <https://handbook.fca.org.uk/handbook/SYSC/25/9.html>
- I18** · FINRA Notice 08-18 and NY DFS letters: mandatory ~2-week block leave with access cut; finance's compulsory bus-factor test. <https://www.finra.org/rules-guidance/notices/08-18>
- I19** · Jabrayilzade et al., "Bus Factor In Practice" (ICSE-SEIP 2022): 75% rate it important, 19% saw it measured; six person-months lost to one departure. <https://arxiv.org/pdf/2202.01523>

G · AI frontier and risk quantification

- G1** · Edge et al., Microsoft GraphRAG: entity graph + community summaries for global questions over a corpus; open source. <https://arxiv.org/abs/2404.16130>
- G2** · SWE-QA: repository-level code QA benchmark (intention understanding, cross-file reasoning). <https://arxiv.org/abs/2509.14635>
- G3** · CoMRAT: detects decision and rationale sentences in commit messages. <https://arxiv.org/html/2506.10986>
- G4** · Design-rationale mining as an established field (text-mining rationale; ASE 2022 rationale management). <https://dl.acm.org/doi/10.1145/3551349.3559568>
- G6** · Park et al., "Generative Agent Simulations of 1,000 People": two-hour interviews; ~85% of subjects' own test-retest consistency. <https://arxiv.org/abs/2411.10109>
- G7** · Rajaraman et al. (NeurIPS 2020): fundamental limits of imitation learning; clone error compounds off-distribution. <https://par.nsf.gov/servlets/purl/10237319>
- G8** · FlowHFT: imitation learning of trading policies; regime fragility is the known weakness. <https://arxiv.org/abs/2505.05784>
- G9** · Delphi ("digital minds"): shipping expert-clone product. <https://www.fastcompany.com/91356476/delphi-ai-digital-mind>
- G10** · The knowledge acquisition bottleneck (Expert Systems, 1988; Cooke 1994): the named 40-year-old problem. <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-0394.1988.tb00065.x>
- G11** · LLMREI: LLM elicitation interviewer matches human interviewer error rates. <https://arxiv.org/abs/2507.02564>
- G12** · AutoElicit (ICML 2025): elicited knowledge as Bayesian priors; fewer labels, lower error. Demonstrated outside finance. <https://arxiv.org/abs/2411.17284>
- G13** · Settles and Meeder (ACL 2016): half-life regression for spaced repetition, deployed at Duolingo. <https://aclanthology.org/P16-1174/>
- G14** · Morgan Stanley x OpenAI: assistant used by >98% of advisor teams; "Debrief" meeting agent shipped. <https://openai.com/index/morgan-stanley/>
- G15** · BloombergGPT: in-house finance LLM on proprietary data. <https://arxiv.org/abs/2303.17564>
- G16** · JPMorgan LLM Suite: internal LLM portal for ~140-200K employees. <https://www.cnbc.com/2024/08/09/jpmorgan-chase-ai-artificial-intelligence-assistant-chatgpt-openai.html>
- G17** · RFC 3161 trusted timestamping; ANSI X9.95 financial-grade extension. <https://datatracker.ietf.org/doc/html/rfc3161>
- G18** · Tamper-evident Merkle logs (Crosby-Wallach; Google Trillian / Certificate Transparency infrastructure). <https://transparency.dev/verifiable-data-structures/>
- G19** · OpenTimestamps git integration: free cryptographic anchoring of commit hashes. <https://github.com/opentimestamps/opentimestamps-client/blob/master/doc/git-integration.md>
- G20** · Instructional fingerprinting of LLMs (NAACL 2024): provable model ownership after fine-tuning. <https://arxiv.org/abs/2401.12255>
- G21** · Avelino et al.: truck factor computed from git history; 46% of popular projects at TF=1; open-source tool. <https://peerj.com/preprints/1233/>
- G22** · Nassif and Robillard (ICSME 2017): turnover-induced knowledge loss as rare-severe-event risk. <https://www.cs.mcgill.ca/~martin/papers/icsme2017.pdf>

- G23** · Bird et al. (FSE 2011, Microsoft): ownership concentration predicts defects. <https://cabird.com/pdfs/bird2011dtm.pdf>
- G24** · Mockus and Herbsleb (ICSE 2002): Expertise Browser, who-knows-what from change history. <https://herbsleb.org/web-pubs/pdfs/mockus-expertise-2002.pdf>

L · Legal and regulatory

- L1** · Winston and Strawn, reasonable-measures primer: ~11% of trade-secret cases dismissed on reasonable measures; case list. <https://www.winston.com/en/insights-news/reasonable-measures-for-protecting-trade-secrets-a-primer>
- L2** · IPWatchdog (2026), reasonable particularity under the DTSA: claims dismissed for vague identification "regardless of whether actual misappropriation occurred" (NEXT v. CLEAResult; Syntel v. TriZetto won on documentation). <https://ipwatchdog.com/2026/01/28/identifying-trade-secrets-under-dtsa-critical-requirement-reasonable-particularity/>
- L4** · Asia IP on Jane Street v. Millennium: attorney prescription verbatim, "document development and implementation meticulously." <https://asiaiplaw.com/section/in-depth/can-a-trading-strategy-be-protected-as-a-trade-secret>
- L5** · DOJ SDNY (2025): trader arrested for stealing a quant firm's "alphas" developed at a cost exceeding \$1B; theft is now criminal enforcement. <https://www.justice.gov/usao-sdny/pr/trader-arrested-stealing-trade-secrets-global-quantitative-trading-firm>
- L6** · MiFID II RTS 6 (EU 2017/589), Art. 5(7): record when, who made, who approved, and the nature of every material algo change; Art. 9: annual self-assessment and validation report. <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32017R0589>
- L7** · FCA 2018 algo review: key decisions taken "without an adequate audit trail and without sufficient explanation of the underlying rationale." <https://www.fca.org.uk/publication/multi-firm-reviews/algorithmic-trading-compliance-wholesale-markets.pdf>
- L9** · Fed SR 11-7: documentation detailed enough for "parties unfamiliar with a model to understand how the model operates, as well as its limitations and key assumptions." <https://www.federalreserve.gov/supervisionreg/srletters/sr1107.htm>
- L10** · SEC Advisers Act Rule 204-2: written communications relating to recommendations and advice are required records. <https://www.ecfr.gov/current/title-17/chapter-II/part-275/section-275.204-2>
- L12** · Holland and Knight: \$3B+ in penalties, 100+ firms, for failing to preserve business communications since 2021. <https://www.hklaw.com/en/insights/publications/2024/12/a-long-winters-nap-sec-off-channel-communications>
- L13** · FCA SYSC 10A: mandatory recording including intent-stage communications; five-to-seven-year retention. <https://handbook.fca.org.uk/handbook/sysc10a>
- L14** · ICO monitoring-workers guidance (2023): lawful basis, proportionality, transparency, DPIA; the design checklist. <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/employment/monitoring-workers/>
- L15** · German BetrVG 87(1)(6): works-council co-determination over monitoring technology; architect as work-product capture. <https://www.employee-monitoring.net/compliance/employee-monitoring-laws-germany>
- L16** · ILPA Principles 3.0: key-person event should trigger automatic suspension of the investment period; succession planning demanded. https://ilpa.org/wp-content/uploads/2019/06/ILPA-Principles-3.0_2019.pdf
- L17** · Capco study (via ODD literature): ~50% of hedge fund failures are operational, vs 38% investment. [https://en.wikipedia.org/wiki/Operational_due_diligence_\(alternative_investments\)](https://en.wikipedia.org/wiki/Operational_due_diligence_(alternative_investments))
- L18** · Buy-over-build at the top: Global Relay serves 22 of the top 25 banks; Smarsh 18 of the top 20. <https://www.businesswire.com/news/home/20250113298920/en/Global-Relay-placed-in-2025-Gartner-Magic-Quadrant-for-Digital-Communications-Governance-and-Archiving-Solutions>
- L19** · SEC Rule 17a-4(f): tamper-evident storage plus a designated third party able to deliver records to regulators. <https://www.sec.gov/rules-regulations/staff-guidance/trading-markets-frequently-asked-questions/rule-amendments-broker>
- L20** · Federal Rule of Evidence 803(6): contemporaneous, regular-practice business records are admissible; the evidentiary payoff of ambient capture. https://www.law.cornell.edu/rules/fre/rule_803

Integrity notes: the Stanford twin result (G6) is on social-survey responses, not trading decisions; always present it with the imitation-learning limit (G7). The claim that pod shops make documentation a condition of capital allocation could NOT be verified publicly; do not assert it, use the risk-gating analogy plus SYSC

25.9 and AIMA DDQs instead. Bridgewater material (I9, I10) is a cultural precedent and a cautionary tale in one; cite both halves. Paywalled or abstract-only entries: I12, G10, and the HBS case (I9).

From Research to Product: What We Ship

This part condenses the product scope the research above produced. The full engineering contract lives in the repo (prd.md, design.md, masterplan.md, docs/scoping.md); this is the judge-and-reader version.

1 · The product in one paragraph

Continuity is a database with an opinion: an append-only, hash-chained decision ledger for trading-strategy research, fed by ambient capture (a git hook that drafts decision records from diffs, a desktop quick-capture hotkey, an LLM debrief agent whose every question is grounded in a captured artifact), read by knowledge-risk analytics (bus factor, concentration, a vacation-readiness score modeled on the regulator-mandated two-week block leave), and generating handover packs shaped to the FCA's own handover rule, RTS 6 style change logs, and cited answers to plain-language questions. Decision tagging runs on a small model we fine-tune ourselves on-prem with the MLX LoRA recipe proven in our prior distillation project, because this data can never leave the building. The ledger head is anchored to Bitcoin via OpenTimestamps, making the record tamper-evident and third-party datable: the reasonable measures and reasonable particularity that trade-secret courts demand.

2 · The schema is the product

ENTITY	WHAT IT HOLDS
events	The ledger. Append-only (UPDATE/DELETE revoked plus a guard trigger); every row carries prev_hash and this_hash computed in-database under a per-firm advisory lock; everything else in the system is a projection of this table
strategies, artifacts	The unit of continuity, and the captured exhaust (commits, notebooks, parameter files) hanging off it, content-hashed and immutable
decisions, decision_links	The genealogy: what changed, why, what was rejected, typed and risk-flagged by the on-prem tagger; edges for supersedes, informs, reverts; machine drafts are visibly drafts until a human approves
debrief_sessions, turns, questions	The interviewer corpus and the ranked feed of open questions only one person can answer
knowledge_scores	Bus factor, concentration, vacation-readiness per strategy, recomputed as the corpus changes
anchor_receipts, handover_packs	OpenTimestamps receipts for ledger heads; frozen generated packs with their ledger position

3 · Total context, by default

The vision layer above the schema: maximum context, continuously captured. The trading floor is already the most-recorded room in the economy (regulators have fined firms more than \$3B for failing to capture business communications, and trader-voice transcription is installed infrastructure), yet that record is write-only: it exists for surveillance, never for memory. Continuity treats total context as the ambient default. Meetings are first-class artifacts: recorded, transcribed, speaker-tagged, filed into the ledger, and citable by decisions, debriefs, and answers. The weekend build ships this as seeded transcripts plus a transcript importer; the

roadmap is always-on desk capture devices that transcribe on-prem and file straight into the ledger, plus taps into the compliance archives firms already maintain, so context capture happens all the time without anyone doing anything. The doctrine underneath it: capture runs continuously, 24/7, as ordinary practice rather than an act anyone performs, and it tracks the work, never the worker. No screen or keystroke monitoring, no behavioral scoring; capture attaches to artifacts, authorship is recorded because provenance is the legal and operational value, and the analytics layer is deliberately de-personalized: risk scores are properties of strategies, never league tables of people. The boundary travels with the vision: work product only, transparent to the people it protects, inside the legal envelope this dossier maps (mandatory-recording rules on one side, monitoring law and works-council constraints on the other).

The governance architecture borrows deliberately from Palantir, the company that proved organizations pay nine figures for an integration-and-memory layer over their own data. Three of its patterns are adopted directly: everything tethered to source (every decision links its provenance, the way Foundry tethers every datum through lineage); access as an audited event (opening or exporting a record appends to the same ledger, with a stated justification on exports, Palantir's checkpoint pattern and its most durable answer to critics); and the processor stance (deployed into the firm's environment, single tenant, the vendor never pooling or training on customer data). One pattern is adopted as a warning: purpose creep, not capture, is what detonates products in this lane, so allowed uses (continuity, onboarding, compliance, IP documentation) are hard-scoped and forbidden uses (performance management, individual productivity analytics) are made technically awkward by construction. The direct precedent is instructive: Palantir and Credit Suisse built a joint venture, Signac, to monitor traders in 2016; it died in fifteen months for governance reasons, not technical ones, and nobody has productized the lane since. Full research notes with sources: docs/palantir.md in the repository.

4 · The demo arc

One story, six beats: a researcher approves a machine-drafted decision record in one keystroke and it flies into the ledger; the on-prem tagger labels it; a resignation triggers the departure simulation and the knowledge map redders; the exit debrief and the ask-bar prove the strategy now answers questions its author is no longer present to answer; compliance exports the change log; and the verify sweep catches a staged tamper on the exact row, with a Bitcoin-anchored receipt on screen. The film script is written to this arc.

5 · Form factor, stack, and deployment

One React frontend, two shells: a Tauri 2 desktop app (menu-bar presence, global-hotkey quick capture, unsigned dmg for the demo machine) and the same frontend deployed on Vercel behind real email-and-password auth, seeded with a synthetic firm (Meridian Basis Partners, five personas) that visitors can extend; all data synthetic by construction and labelled as such. Supabase carries Postgres, auth, row-level security by firm, realtime ledger tail, and pgvector retrieval for cited answers. Every version and API in the plan was verified against live registries during the research stage, down to the capability-file traps and the fact that Bitcoin attestation takes hours, so the first ledger head gets stamped on day one. The build stage executes a sprint ledger with budgets, acceptance blocks, pre-agreed fallbacks, and a logging rule that records completion time and status at the close of every sprint.

6 · Honesty commitments (carried from the research)

No number appears in the product, README, or pitch that a committed artifact cannot back. The tagger quotes its own held-out evaluation or the clearly-attributed fallback. Machine drafts are visibly machine drafts. The twin stays a pitch beat, not a product claim. The Jane Street figure is always attributed to the firm's court filing. Amber appears in the interface only where knowledge risk is being signalled.